

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

1. Explain the difference between the function $f(x) = |x|$ and the function $g(x) = \sqrt{x^2}$.
2. Compute the following expressions. You should use the unit circle or special triangles.
 - $\sin\left(\frac{3\pi}{4}\right)$.
 - $\tan\left(\frac{7\pi}{6}\right)$.
 - $\sin\left(\frac{7\pi}{3}\right)$.
 - $f(g(\pi))$ where $f(x) = \sqrt{x+1}$ and $g(x) = \cos(x)$.

Exercise 1.

For each of the following functions $h(x)$, find a function $f(x)$ and a function $g(x)$ such that $h(x)$ is the composite function $f(g(x))$.

1. $h(x) = \sin(3^x)$.
2. $h(x) = 2^{x^2}$.
3. $h(x) = |\cos(x)|^2$.

Exercise 2.

1. Sketch the graph of $\sin(x)$.
2. Sketch the graph of $\frac{1}{x}$.
3. Sketch the graph of $\sin\left(\frac{1}{x}\right)$.
4. Give the domain of $\sin\left(\frac{1}{x}\right)$.
5. Give the range of $\sin\left(\frac{1}{x}\right)$.